

Disproof of Conjecture 00000001195 (Kronecker power-sum “orthogonality”)

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2026-10-05

1 The conjecture

English text (verbatim apart from Unicode): *Definition: The inner (Kronecker) product of symmetric functions. Conjecture: $\langle p_\lambda, p_\mu * p_\nu \rangle = z_\lambda \cdot \delta$ with lexicographic counting (Kronecker power-sum orthogonality).* The Chinese text says the same: the definition line names the internal product (Kronecker) of symmetric functions, and the conjecture is $\langle p_\lambda, p_\mu * p_\nu \rangle = z_\lambda \delta$ (“Kronecker power-sum orthogonality”).

Reading. p_λ is the power-sum symmetric function, $\langle \cdot, \cdot \rangle$ is the Hall inner product, $*$ is the Kronecker (internal) product named in the definition line, $z_\lambda = \prod_{i \geq 1} i^{m_i} m_i!$ where m_i is the number of parts of λ equal to i , and δ is a Kronecker delta, i.e. a quantity with value 0 or 1 (the natural choice being $\delta_{\lambda\mu} \delta_{\mu\nu}$). The claim is that the identity holds for all partitions λ, μ, ν of a common n .

Result. At $\lambda = \mu = \nu = (2)$ we prove $\langle p_{(2)}, p_{(2)} * p_{(2)} \rangle = 4 = z_{(2)}^2$, while $z_{(2)} = 2$, so $z_{(2)} \cdot \delta \in \{0, 2\}$ for every $\delta \in \{0, 1\}$. The conjecture is therefore false for every choice of a 0/1-valued δ . (The correct identity is $\langle p_\lambda, p_\mu * p_\nu \rangle = z_\lambda^2 \delta_{\lambda\mu} \delta_{\mu\nu}$; only the instance $\lambda = (2)$ is used and formalized.)

Readings not covered. (i) If $*$ were the ordinary product of symmetric functions, the formula $\langle p_\lambda, p_\mu p_\nu \rangle = z_\lambda \delta_{\lambda, \mu \cup \nu}$ is true; we do not use this reading because the text defines $*$ as the inner (Kronecker) product. (ii) If “ δ with lexicographic counting” were meant as some integer count other than a 0/1 delta, the statement does not say which; such readings are not addressed.

2 Definitions and conventions

Two variables suffice. Wikipedia, *Ring of symmetric functions* (https://en.wikipedia.org/wiki/Ring_of_symmetric_functions): “It suffices to take for d the degree of the symmetric function, since the part in degree d of that ring is mapped isomorphically to rings with more indeterminates by” [the restriction maps] “This implies that for studying relations between individual elements, there is no fundamental difference between symmetric polynomials and symmetric functions.” We therefore work with the degree-2 part of $\mathbb{Q}[x_0, x_1]^{S_2}$, inside Mathlib’s `MvPolynomial (Fin 2) Q`. The bases are Mathlib’s: $p_\lambda = \text{psumPart}$, $h_\lambda = \text{hsymmPart}$, $m_\lambda = \text{msymm}$. Explicitly (lemmas `psumPart_two`, `hsymmPart_two`, `hsymmPart_oneOne`, `msymm_two`, `msymm_oneOne`): $p_{(2)} = x_0^2 + x_1^2 = m_{(2)}$, $h_{(2)} = x_0^2 + x_0 x_1 + x_1^2$, $h_{(1,1)} = (x_0 + x_1)^2$, $m_{(1,1)} = x_0 x_1$. The partitions (2) and $(1, 1)$ are the only partitions of 2 (`partition_two_cases`).

z_λ . `zee` $\lambda = \prod_i i^{m_i} m_i!$, the product over the *distinct* part sizes i of λ (Lean: `1.parts.toFinset`), not over the parts counted with multiplicity. Wikipedia, *Kronecker coefficient* (https://en.wikipedia.org/wiki/Kronecker_coefficient) gives the same coefficients as $z_\rho = \prod_j j^{i_j} i_j! = n!/|C_\rho|$, C_ρ the conjugacy class of cycle type ρ . Here $z_{(2)} = 2$ (`zee_two`).

Hall inner product. A bilinear form B is a Hall inner product on the degree-2 part if $B(h_\lambda, m_\mu) = \delta_{\lambda\mu}$ for all $\lambda, \mu \vdash 2$ (`IsHallInner`). Since $\{h_\lambda\}$ and $\{m_\mu\}$ are bases of the degree-2 part, this determines B there.

Frobenius characteristic and Kronecker product. For $\varphi : S_2 \rightarrow \mathbb{Q}$ put

$$\text{ch}(\varphi) = \frac{1}{2!} \sum_{w \in S_2} \varphi(w) p_{\rho(w)}$$

(`frob`), where $\rho(w)$ is the cycle type of w with fixed points counted as parts 1 (Mathlib's `Equiv.Perm.partition`). Grouping by classes and using $z_\rho = n!/|C_\rho|$ gives $\text{ch}(\varphi) = \sum_\rho z_\rho^{-1} \varphi(\rho) p_\rho$, and Wikipedia, *Schur polynomial* (https://en.wikipedia.org/wiki/Schur_polynomial) gives $s_\lambda = \sum_\nu \frac{\chi_\nu^\lambda}{z_\nu} p_\nu$, i.e. $s_\lambda = \text{ch}(\chi^\lambda)$. Wikipedia, *Kronecker coefficient*: “Kronecker coefficients $g_{\lambda\mu\nu}$ describe the decomposition of the tensor product (= Kronecker product) of two irreducible representations of a symmetric group into irreducible representations”, and “One can interpret this on the level of symmetric functions, giving a formula for the Kronecker product of two Schur polynomials”, namely $s_\mu \star s_\nu = \sum_\lambda g_{\mu\nu}^\lambda s_\lambda$. Since the character of a tensor product is the pointwise product of characters, $\chi^\mu \chi^\nu = \sum_\lambda g_{\mu\nu}^\lambda \chi^\lambda$; applying the linear map ch gives $\text{ch}(\chi^\mu) \star \text{ch}(\chi^\nu) = \text{ch}(\chi^\mu \chi^\nu)$, and by bilinearity

$$\text{ch}(\varphi) \star \text{ch}(\psi) = \text{ch}(\varphi\psi) \quad (\varphi, \psi \text{ class functions}).$$

A bilinear map K is a Kronecker product on the degree-2 part if it satisfies this for all class functions φ, ψ on S_2 (`IsClassFun`, `IsKronecker`). As ch maps onto the degree-2 part, this determines K there.

3 Proof

Lemma 1 (`hall_p2_p2`). *For every Hall inner product B , $B(p_{(2)}, p_{(2)}) = 2$.*

Proof. $p_{(2)} = x_0^2 + x_1^2 = 2h_{(2)} - h_{(1,1)}$ (`psumPart_two_eq_h`) and $p_{(2)} = m_{(2)}$, so $B(p_{(2)}, p_{(2)}) = 2B(h_{(2)}, m_{(2)}) - B(h_{(1,1)}, m_{(2)}) = 2 - 0$. $\square$

Lemma 2 (`frob_eq`, `frob_classInd_two`, `kron_p2_p2`). $\text{ch}(\varphi) = \frac{1}{2}(\varphi(1)(x_0 + x_1)^2 + \varphi(s)(x_0^2 + x_1^2))$ with $s = (01)$. With $\varphi_2 = z_{(2)} \cdot \mathbf{1}_{\{\rho(w)=(2)\}}$ (`classInd_two`, a class function by `classInd_isClassFun`) we have $\text{ch}(\varphi_2) = p_{(2)}$. Hence for every Kronecker product K , $K(p_{(2)}, p_{(2)}) = \text{ch}(\varphi_2^2) = \text{ch}(2\varphi_2) = 2p_{(2)}$.

Proof. $S_2 = \{1, s\}$, $\rho(1) = (1,1)$, $\rho(s) = (2)$ (`univ_perm_two`, `parts_partition_one`, `parts_partition_swap`). $\varphi_2(1) = 0$, $\varphi_2(s) = 2$, so $\text{ch}(\varphi_2) = x_0^2 + x_1^2$; $\varphi_2^2 = 2\varphi_2$ pointwise, and ch is linear (`frob_smul`). $\square$

Lemma 3 (existence: `B0_isHall`, `K0_isKronecker`). *Let $a(f) = f(1,0)$, $b(f) = f(1,1) - 2f(1,0)$ (on the degree-2 part these are the coordinates of f in the basis $m_{(2)}, m_{(1,1)}$). Then $B_0(f, g) = (2a(f) - b(f))a(g) + (b(f) - a(f))b(g)$ is a Hall inner product and $K_0(F, G) = \frac{1}{2}(b(F)b(G)(x_0 + x_1)^2 + (2a(F) - b(F))(2a(G) - b(G))(x_0^2 + x_1^2))$ is a Kronecker product.*

Proof. Direct evaluation on $h_{(2)}, h_{(1,1)}, m_{(2)}, m_{(1,1)}$; and $a(\text{ch}\varphi) = (\varphi(1) + \varphi(s))/2$, $b(\text{ch}\varphi) = \varphi(1)$ ($\text{ca_frob}, \text{cb_frob}$), so $K_0(\text{ch}\varphi, \text{ch}\psi) = \text{ch}(\varphi\psi)$. $\square$

Theorem 4 (`conjecture_1195_false`). *Hall inner products and Kronecker products exist, and for all of them $\langle p_{(2)}, p_{(2)} * p_{(2)} \rangle = 4 = z_{(2)}^2$, $z_{(2)} = 2$, and $\langle p_{(2)}, p_{(2)} * p_{(2)} \rangle \neq z_{(2)} \cdot \delta$ for $\delta \in \{0, 1\}$.*

Proof. $B(p_{(2)}, K(p_{(2)}, p_{(2)})) = B(p_{(2)}, 2p_{(2)}) = 2 \cdot 2 = 4$, while $z_{(2)}\delta \in \{0, 2\}$. $\square$

4 Lean formalization

File `lean/Conjecture1195/Basic.lean` (273 lines, only `import Mathlib`, Lean v4.33.1, Mathlib v4.33.1). Main statement:

```
theorem conjecture_1195_false :
  (exists B, IsHallInner B) /\ (exists K, IsKronecker K) /\
  forall B K, IsHallInner B -> IsKronecker K ->
    B (psumPart (Fin 2) Q two) (K (psumPart (Fin 2) Q two) (psumPart (Fin 2) Q two))
      = (zee two : Q) ^ 2 /\ zee two = 2 /\
    forall d : Q, (d = 0 \/ d = 1) ->
      B (psumPart (Fin 2) Q two) (K (psumPart (Fin 2) Q two) (psumPart (Fin 2) Q two))
        /= (zee two : Q) * d
```

(written here in ASCII; B and K are $\mathbb{Q}$ -bilinear maps on `MvPolynomial (Fin 2) Q`). `lake build` succeeds with no warnings; `#print axioms C1195.conjecture_1195_false` reports only `propext`, `Classical.choice`, `Quot.sound`. No sorry, no `native_decide`; `decide` is used only for finite enumerations of S_2 , of $\text{Sym}^2(\text{Fin } 2)$ and of multiset (in)equalities.